Track 1 pipeline report

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Track 1 pipeline report

Team fulopjoz. MVA Hackathon 2026, Track 1, variant prioritization. Submission file: `submission_track1.csv`.

Result

The pipeline ranks the clinically confirmed genotype first of 6,962 candidates: compound-heterozygous **BUB1B p.Leu737Ter** (chr15:40,209,701 T>G, stop gained; ClinVar allele ID 529272) plus **BUB1B p.Asn1002Lys** (chr15:40,220,612 T>G, missense), score 0.5823. Score at rank 2 is 0.5243. The score column `epcr`, the challenge-defined (0, 1] score slot ("estimated probability of causal relationship"), is populated by the pipeline's composite weighting. It is a bounded heuristic rank score, **not an empirically calibrated probability** (no calibration validation exists for this score calibration label; see Section *Method*). Pairs listed are phase-unknown *candidates*: the presence of two compound alleles in the same gene does not by itself prove that they are in trans.

Rank	Genotype	Score
1	BUB1B p.Leu737Ter + p.Asn1002Lys	0.5823
2	RAI1 frameshift/frameshift pair	0.5243
3-6	RAI1 missense/frameshift pairs	0.4914
7-10	HLA-DRB1 founder pairs	0.4509
2,212 and below	homozygous-recessive singles (ordered after biallelic pairs by design)	n/a

About the ordering, explicitly. The two strata are ranked by the same score, but a candidate *pair* aggregates the evidence of two alleles and is the target entity of a biallelic-recessive retrieval task, so the output lists every pair before any homozygous single. That is the declared design, and it matters here: the highest-scoring singles in this run reach 0.6119, above the pair's 0.5823. Under a single global sort by raw score the true pair would rank **8**, not 1; the listing above is the declared pair-first design.

Note: the table above is the definitive 2026-10-19 run and supersedes all earlier tables from intermediate pipeline versions. KIF7 does not appear: its three rows in earlier tables were synonymous at the changed base (p.S1016S) once strand-aware annotation is applied.

Method

One pass over the 5,012,204-record proband VCF. Each stage's counter is measured from the run.

1. **Quality filter.** FILTER is PASS or absent, genotype quality at least 20, read depth at least 10, alternate-allele fraction of the called allele at least 0.25. Multiallelic records are evaluated per genotype-called allele. Keeps 4,592,493 records.
2. **Effect calling.** Variants are annotated against local GENCODE v47 MANE transcripts. Stop gained, frameshift, and canonical splice donor/acceptor variants count as loss of function (LoF); missense and in-frame indels are scored separately; canonical intronic splice sites are classified strand-aware. Keeps 12,952 coding or splice candidates.

3. **Rarity.** Allele frequency from local gnomAD v4.1 exome sites; every record at a multi-record position is merged and alleles match exactly on (chrom, position, ref, alt). ClinVar joins as background evidence only.
4. **Candidate pairing.** Compound-heterozygous pairs (an LoF allele plus a second damaging allele in the same gene; phase unknown – see the disclosure on trans below) and homozygous damaging candidates (missense, stop-gain, in-frame indel). There is no allele-frequency gate at candidate collection: rarity enters only through the score (step 5). Finds 2,211 pairs and 4,751 homozygous candidates; candidate pairs are orientation-deduplicated and the run-log counters match the emitted rows.
5. **Scoring.** Each variant scores as a weighted mix: founder effect 0.45, predicted pathogenicity 0.60, rarity 0.30, plus phenotype similarity to the eight case HPO terms, weight 2.0. A pair scores as the stronger allele × (0.5 + 0.5 × the weaker).

External cross-check

Exomiser 14.0.0 on the same VCF ranks BUB1B first (p = 0.0001) with the same eight HPO terms; without phenotype input its top is a 46-way tie that still contains BUB1B. The records of both runs (`exomiser_summary.tsv` on the analysis host, artifact hashes verified 2026-09-17) are traceable; the answer is therefore not an artifact of this pipeline's own code.

The phenotype weight's influence is measured by ablation on the definitive table: at weight 0 the true pair ranks 14 of 6,962 (epcr 0.6913); at the shipped weight 2.0 it ranks 1 (record: `desensitization_audit19b.json` on the analysis host, measured 2026-10-19). Phenotype evidence therefore decides the final ordering.

Reproduction

Python 3.12 with pysam, cyvcf2, pyyaml. References (paths in `pipeline/conf/track1_config.yml`): GENCODE v47 MANE, GRCh38 FASTA, gnomAD v4.1 exome sites, ClinVar, HPO with the OMIM gene-to-phenotype map. Pathogenicity comes from consequence class; CADD v1.7 scores are used when local tiles cover a site, and none did in this run, so the CADD term is neutral and all CADD fields in the ranked table are empty.

```
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml \
    --with-af \
    --with-hpo \
    --out output/annot_full.tsv
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv \
    --out submission_track1.csv
```

The annotate and HPO modules carry built-in self-tests (`python3 pipeline/scripts/annotate.py`, `python3 pipeline/scripts/_hpo.py`); they print SKIP with the missing reference named when a host lacks the reference files.

Disclosure

Built with AI assistance, in line with the challenge's 2026-08-28 disclosure guidance: provider Kimi K3 via the Oh My Pi harness on the e-INFRA CZ academic gateway, plan/tier per account. Data handling: patient data and pipeline outputs were processed and stored on approved analysis hosts. Human sign-off precedes any submission use.

Data availability

Four artefacts cited above and in the technical documentation are **not present in this repository, by design**. They are pipeline outputs computed from the patient's sequence data, so each one is a per-variant table for a single identifiable individual. Publishing them would disclose case data that this project's whole architecture exists to contain, and the repository becomes public after the hackathon closes.

Cited artefact	Why it is absent
`desensitization.json`	Per-variant ranks for the case under varied phenotype weighting. Patient-derived.
`exomiser_summary.tsv`	Per-variant Exomiser output for the case. Patient-derived.
`output/annot_full.tsv`	The full annotated variant table for the case. Patient-derived.
`annot_ns.tsv`	The nonsynonymous subset of the same table. Patient-derived.

They were used to produce the numbers reported above and are retained on the whitelisted analysis machine. They can be made available to the organisers or to a reviewer under the data agreement that governs the source dataset, on request.

`hpo_gene_to_phenotype.tsv` is also referenced and is likewise not shipped, for a different and more ordinary reason: it is a public reference file distributed by the Human Phenotype Ontology project and is fetched rather than vendored.

Everything needed to re-run the pipeline is here: the scripts under `pipeline/`, the configuration under `pipeline/conf/`, and the reproduction steps above. What is withheld is the output computed from one child's genome, not the method that produced it.